

```

import pandas as pd
#df = pd.read_csv('seaice.csv')

from google.colab import drive
drive.mount('/content/drive')
# 데이터 불러오기
df = pd.read_csv('/content/drive/My Drive/소프트웨어와생활/seaice.csv',
                  encoding='cp949'
)

# ① 열 이름 앞뒤의 공백 제거
df.columns = df.columns.str.strip()

# ② 결측치 확인
print(df.isnull().sum())

# ③ 중복 데이터 확인
print(df.duplicated().sum())

# ④ Month와 hemisphere의 값 확인
print(df['Month'].unique())
print(df['hemisphere'].unique())

print(df.columns)
print(list(df.columns))
print(df[['Year', 'Month', 'Day', 'Extent', 'hemisphere']])

```

```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.r
Year      0
Month     0
Day       0
Extent    0
Missing   0
Source Data 0
hemisphere 0
dtype: int64
0
[10 11 12  1  2  3  4  5  6  7  8  9]
['north' 'south']
Index(['Year', 'Month', 'Day', 'Extent', 'Missing', 'Source Data',
      'hemisphere'],
      dtype='object')
['Year', 'Month', 'Day', 'Extent', 'Missing', 'Source Data', 'hemisphere']

```

	Year	Month	Day	Extent	hemisphere
0	1978	10	26	10.231	north
1	1978	10	28	10.420	north
2	1978	10	30	10.557	north
3	1978	11	1	10.670	north
4	1978	11	3	10.777	north
...	...	...	...	...	...
26349	2019	5	27	10.085	south
26350	2019	5	28	10.078	south
26351	2019	5	29	10.219	south
26352	2019	5	30	10.363	south
26353	2019	5	31	10.436	south

[26354 rows x 5 columns]

전처리 전 시각화

```
# 한글 폰트 설치
!pip install koreanize-matplotlib
```

```
Requirement already satisfied: koreanize-matplotlib in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: numpy>=1.23 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages
```

```
import pandas as pd
import matplotlib.pyplot as plt
import koreanize_matplotlib

# 날짜 만들기
df['Date'] = pd.to_datetime(
    df[['Year', 'Month', 'Day']]
)
print(df['Date'])
# 전처리 전 일별 원본 데이터 시각화
plt.figure(figsize=(14, 6))

plt.plot(
    df['Date'],
    df['Extent'],
    linewidth=1
)

plt.title('북극 해빙 면적: 일별 원본 시각화', fontsize=18)
plt.xlabel('연도')
plt.ylabel('해빙 면적(백만 km2)')

plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

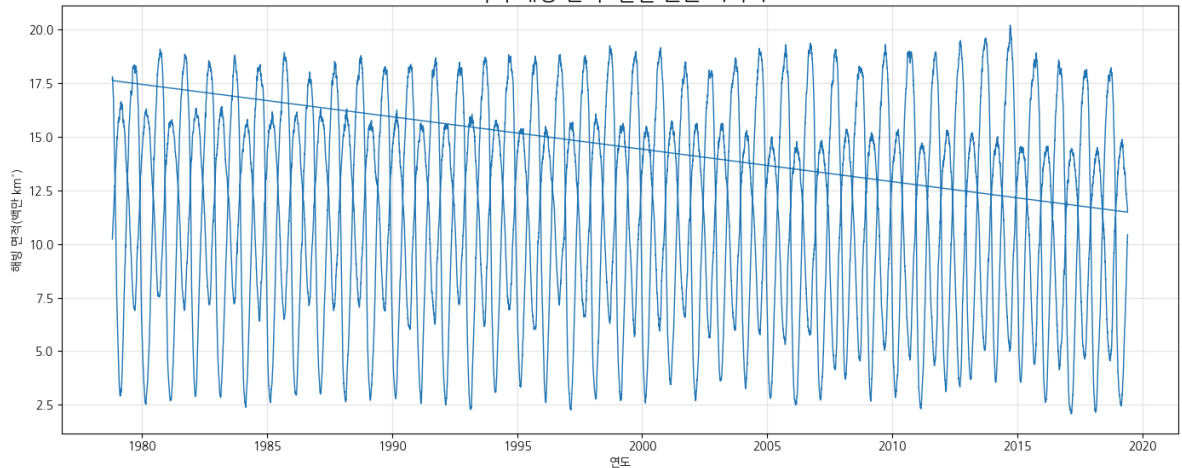
```
0      1978-10-26
1      1978-10-28
2      1978-10-30
3      1978-11-01
4      1978-11-03
```

...

```
26349  2019-05-27
26350  2019-05-28
26351  2019-05-29
26352  2019-05-30
26353  2019-05-31
```

Name: Date, Length: 26354, dtype: datetime64[ns]

북극 해빙 면적: 일별 원본 시각화



1. 북극 해빙이 줄고 있다

2. 1년 중 해빙(바다얼음)이 가장 작아지는 9월만 평균으로 요약

⑤ 북반구의 9월 데이터만 추출

```
df = df[(df['hemisphere'] == 'north') & (df['Month'] == 9)]
```

⑥ 분석에 필요한 열만 추출

```
df = df[['Year', 'Extent']]
```

df

	Year	Extent	
155	1979	7.228	
156	1979	7.147	
157	1979	7.037	
158	1979	7.075	
159	1979	7.092	
...	...	...	
12929	2018	4.758	
12930	2018	4.857	
12931	2018	4.867	
12932	2018	4.846	
12933	2018	4.875	

1080 rows × 2 columns

다음 단계:

[df 변수로 코드 생성](#)[New interactive sheet](#)

```

import pandas as pd
import matplotlib.pyplot as plt
import koreanize_matplotlib
# 연도별 9월 평균 빙하 면적 - 같은 연도의 데이터를 묶어서 Extent의 평균을 구
df_year = df.groupby('Year')['Extent'].mean().reset_index()

# 시각화
import matplotlib.pyplot as plt

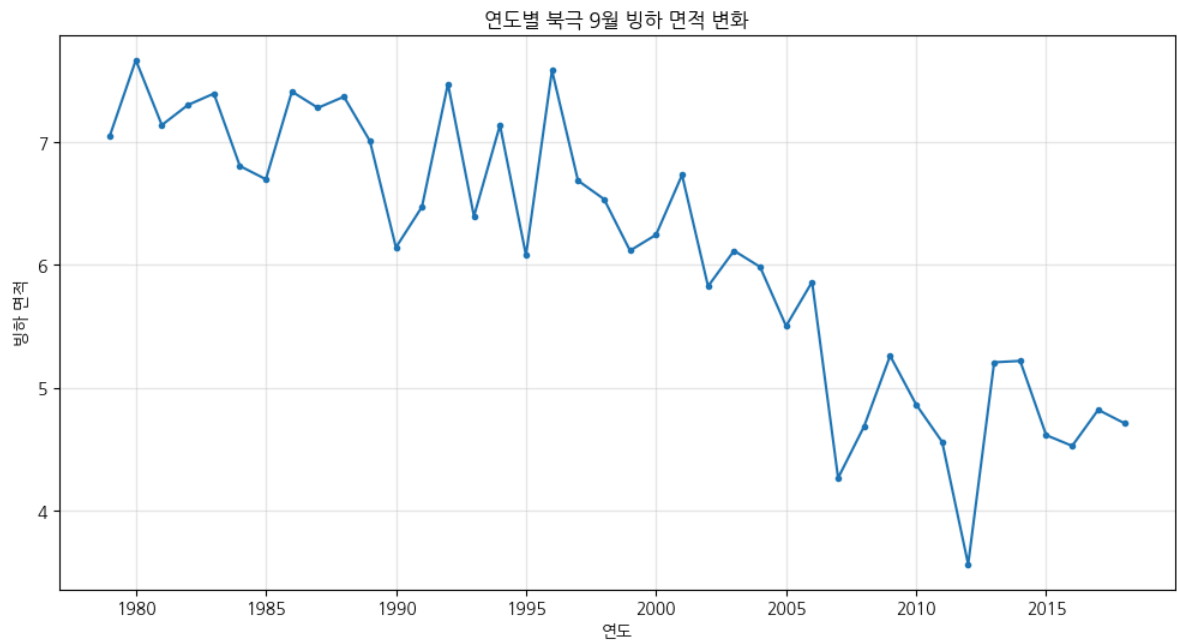
plt.figure(figsize=(12, 6))

plt.plot(df_year['Year'], df_year['Extent'],
         marker='o', markersize=3)

plt.title('연도별 북극 9월 빙하 면적 변화')
plt.xlabel('연도')
plt.ylabel('빙하 면적')

plt.grid(alpha=0.3)
plt.show()

```



```
import numpy as np
#df = pd.read_csv('seaice.csv')

from google.colab import drive
drive.mount('/content/drive')
df = pd.read_csv(
    '/content/drive/MyDrive/소프트웨어와생활/seaice.csv',
    encoding='cp949'
)
# 열 이름 공백 제거
df.columns = df.columns.str.strip()

# 결측치 확인
print(df.isnull().sum())

# 중복값 확인
print(df.duplicated().sum())

# 북반구의 9월 데이터만 추출
df = df[(df['hemisphere'] == 'north') & (df['Month'] == 9)]

# 필요한 열만 추출
df = df[['Year', 'Extent']]

# 전처리 결과 확인
df
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount()
 Year 0
 Month 0
 Day 0
 Extent 0
 Missing 0
 Source Data 0
 hemisphere 0
 dtype: int64
 0

	Year	Extent	
155	1979	7.228	
156	1979	7.147	
157	1979	7.037	
158	1979	7.075	
159	1979	7.092	
...	...	...	
12929	2018	4.758	
12930	2018	4.857	
12931	2018	4.867	
12932	2018	4.846	
12933	2018	4.875	

1080 rows × 2 columns

다음 단계: [df 변수로 코드 생성](#)

[New interactive sheet](#)

```
# 연도별 9월 평균 계산
df_year = df.groupby('Year')['Extent'].mean().reset_index()
print(df_year)

# 그래프
plt.figure(figsize=(12, 6))

plt.plot(df_year['Year'], df_year['Extent'],
         marker='o',
         label='9월 평균 빙하 면적')

# 선형 추세선
기울기, 절편 = np.polyfit(
    df_year['Year'],
    df_year['Extent'],
    1
)
print('='*60)
print(f'기울기: {기울기:.2f} 절편: {절편:.2f}')
```

```
print('='*60)
추세선 = 기울기 * df_year['Year'] + 절편

plt.plot(df_year['Year'], 추세선,
         linestyle='--',
         label='선형 추세선')

plt.title('북극 9월 빙하 면적 변화')
plt.xlabel('연도')
plt.ylabel('빙하 면적')

plt.grid(alpha=0.3)
plt.legend()
plt.show()
```

	Year	Extent
0	1979	7.051133
1	1980	7.667067
2	1981	7.138467
3	1982	7.301733
4	1983	7.394600
5	1984	7.305000